

Training-Free Partial-Order and Distance-Constrained Discrete Diffusion for Controllable POI Check-in Generation

Anonymous Author(s)

Abstract

Location-based social networks (LBSNs) are a core source of web-scale human-mobility data, yet sharing raw check-in traces is privacy-sensitive, motivating realistic synthetic generation. Beyond statistical realism, many downstream web applications need control over the *semantic order* of activities—e.g., leisure before dining—a sample-level, cross-position requirement that single-step conditioning cannot express. We present PCDG, a *training-free*, sampling-time projection layer for conditional joint temporal + category/POI discrete-diffusion generators. From a sample-level category partial-order matrix, PCDG compiles differentiable order-violation and existence energies and, in the late reverse-diffusion steps, projects *only the category-position logits* through a KL-preserving augmented-Lagrangian optimization; a complementary distance-distribution KL term regularizes spatial travel length to preserve geographic fidelity. The projected category logits are then jointly sampled with the unaltered POI distribution. Across two cities under an out-of-distribution split and three sampling seeds, PCDG more than halves the constraint-violation rate and nearly triples pairwise ordering coverage over the unconstrained base generator, surpassing energy guidance, post-hoc reordering, and even a *fully retrained* partial-order conditional model—without any retraining of the base generator. Ablations isolate the role of each energy term, the adaptive multipliers, and the distance regularizer, and we characterize the resulting fidelity–constraint trade-off.

CCS Concepts

• Computing methodologies → Machine learning; • Information systems → Information systems applications.

Keywords

synthetic human mobility, controllable generation, discrete diffusion, constrained sampling, POI check-in sequences, partial-order constraints, LBSN

1 Introduction

Location-based social networks (LBSNs) and location-aware web services generate vast streams of *check-in* data: time-stamped visits to points of interest (POIs), each carrying an activity semantics such as dining, leisure, or shopping. These web-scale traces underpin POI recommendation, urban analytics, and mobility simulation [9, 26, 41]. Yet releasing raw traces is privacy-sensitive, and long-tail scenarios are scarce, which has made the generation of realistic *synthetic* mobility an active research direction [7, 10, 24, 43].

Most prior generators optimize statistical realism alone. Many web applications, however, require *control over activity semantics*—in particular, over the *order* in which categories of activities occur. A scenario designer may need sequences in which leisure precedes dining, or sightseeing precedes shopping, in order to study specific behavioral patterns, stress-test a recommender, or augment data for

a target scenario. Such a requirement is a *sample-level partial order* over categories: it couples multiple, possibly non-adjacent positions of a sequence and therefore cannot be expressed by reweighting any single token, nor by next-location prediction, which outputs one step at a time rather than a complete constrained sequence.

Enforcing such constraints is surprisingly hard for modern generators. Generating freely and then *reordering* the output satisfies the constraint but distorts the jointly learned distribution over time, category, and space. Classifier-free or classifier-based guidance can bias generation toward a condition, but a condition embedding does not *guarantee* a global ordering, and retraining a conditional model for every constraint family is expensive and city-specific. Recent work on controllable trajectory generation enforces *geometric* constraints such as road topology [46], and generic constrained discrete diffusion enforces sequence-level logical constraints [3]; neither has been specialized to the activity-category precedence that arises in POI check-in sequences, nor to the joint temporal + category + POI setting.

We introduce PCDG, a *training-free*, sampling-time projection layer that enforces sample-level category precedence directly inside the reverse diffusion of a frozen conditional generator. PCDG compiles a partial-order matrix into differentiable order-violation and existence energies, and, during the late reverse steps, projects *only the category-position logits* via a KL-preserving augmented-Lagrangian optimization; a complementary distance-distribution regularizer preserves spatial realism. The projected category logits are then jointly sampled with the unaltered POI logits. Because PCDG operates on the per-step sampling distribution of a frozen model, it adds no parameters and needs no retraining, and it can be attached to an existing, already-deployed generator.

Relevance to the Web. This work studies the Web as a socio-economic system and a responsible-data problem. LBSN check-in logs are a native web data source that powers POI recommendation and urban web services, but releasing raw traces risks user privacy. Generating controllable, constraint-satisfying synthetic mobility lets practitioners share and augment web-behavior data while honoring application-specified activity semantics, directly serving privacy-conscious web data sharing and downstream recommendation. The method also contributes a reusable recipe—training-free, sampling-time constraint projection on a frozen generator—for controllable generation of other structured web content.

Contributions.

- We formalize *category partial-order constrained POI check-in generation*—a sample-level, cross-position precedence requirement with an anti-degeneracy existence condition—and express it as differentiable energies compatible with discrete-diffusion sampling (§3, §4.2).

- We propose PCDG, a training-free sampling-time projection that enforces these constraints on category positions via a KL-preserving augmented-Lagrangian optimization and preserves spatial fidelity through a distance-distribution regularizer on POI positions, all on a *frozen* joint temporal + category + POI diffusion generator (§4).
- We conduct a two-city, out-of-distribution, multi-seed study with an eight-way ablation, showing that PCDG attains the strongest constraint satisfaction among energy-guidance, post-hoc, and even *fully retrained* conditional baselines, and we characterize the fidelity–constraint trade-off (§5).

2 Related Work

Human mobility and POI modeling. Classical studies established that human movement is highly regular yet only boundedly predictable [13, 34]. A large body of work predicts the *next* location from history using attentional, spatio-temporal, and graph-enhanced models [9, 26, 41], and diffusion models have recently been applied to next-POI and sequential recommendation [6, 22, 29, 36, 38]. These methods target prediction or ranking of a single next step and do not emit a complete, constraint-satisfying sequence of time-stamped, categorized visits.

Synthetic trajectory and mobility generation. Data-driven generation spans GAN/Seq2Seq models [2, 19, 21, 40], point-process and neural-ODE generators [24, 43], imitation- and privacy-oriented approaches [10, 17, 28, 30, 35, 37], and, increasingly, diffusion models [4, 12, 23, 31, 39, 42, 45] and LLM-based generators [14, 16, 20, 44]. Closest to our backbone, Marionette performs fine-grained conditional generation of spatiotemporal trajectories with a diffusion point process and a category-to-POI cascade [7], and related benchmarks study imitative generation quality [8]. These works pursue distributional realism or coarse conditioning; none enforces a *sample-level partial order over activity categories* within the generated sequence.

Discrete diffusion. Discrete diffusion extends denoising diffusion [15] to categorical data via structured transition matrices [1], concrete score matching [27], score-entropy [25], and masked formulations [32]; guidance mechanisms steer discrete generation toward conditions [33]. We adopt a discrete-diffusion backbone and intervene at sampling time rather than changing training.

Controllable and constrained generation. ControlTraj injects road-topology constraints into trajectory diffusion [46]; Geo-Llama conditions an LLM on mobility constraints [20]. In NLP, structured decoding enforces grammars or DAG automata [5, 11], and precedence constraints are studied in temporal planning [18]. Most relevant methodologically, Constrained Discrete Diffusion (CDD) embeds differentiable constraint optimization into discrete-diffusion sampling via KL projection, Gumbel-softmax relaxation, and augmented Lagrangian updates [3]. PCDG *specializes* this sampling-time projection paradigm to the previously unaddressed case of activity-category precedence in joint temporal + category + POI check-in generation: we contribute the order/existence energy formulation for sample-level partial orders, a category-position-restricted projection coupled with a spatial distance-distribution regularizer on POI positions, and—unlike prior controllable generators that retrain

or post-process—a training-free controller that we show outperforms a *fully retrained* partial-order conditional model on constraint satisfaction (§5).

3 Problem Formulation

POI check-in sequences. We model a user’s daily behavior on an LBSN as a variable-length sequence of check-in events. A sequence of length n consists of arrival times $t_{1:n}$ (continuous, strictly increasing), activity categories $k_{1:n}$ with $k_i \in C = \{1, \dots, C\}$, and points of interest $l_{1:n}$ with $l_i \in \mathcal{L}$. Each POI l carries fixed geographic coordinates $\text{gps}(l)$, and each training POI is associated with a category. A per-sequence context $\mathbf{c}$ encodes coarse temporal and behavioral conditions (e.g., time-of-day occupancy indicators). The empirical distribution over such sequences is highly structured: it reflects the regularity and bounded predictability of human mobility [13, 34] while retaining substantial individual variability.

Category partial-order constraints. Many web applications do not merely require a statistically realistic sequence; they require that the *activity semantics* follow an application-specified order—for instance, that leisure precedes dining, or that sightseeing precedes shopping. We formalize such requirements as a sample-level *partial order* over categories. Given a reference or user-specified ordering, we encode it as a binary matrix $\mathbf{M} \in \{0, 1\}^{C \times C}$, where $\mathbf{M}[a, b] = 1$ denotes the strict precedence constraint “category a precedes category b ”: in a satisfying sequence, *every* occurrence of category a must appear before *every* occurrence of category b . When $\mathbf{M}$ is derived from a reference sequence, we set $\mathbf{M}[a, b] = 1$ iff the last position of a precedes the first position of b . A constraint (a, b) is *violated* either by an inversion (some b before some a) or, degenerately, by dropping a or b entirely so that the ordering is vacuously “satisfied.” A faithful method must avoid both.

Task. Given a frozen conditional generator of complete check-in sequences and a sample-level partial-order matrix $\mathbf{M}$, we seek to sample

$$(t_{1:n}, k_{1:n}, l_{1:n}) \sim p(t_{1:n}, k_{1:n}, l_{1:n} \mid \mathbf{c}, \mathbf{M}), \quad (1)$$

such that (i) the generated sequences remain close to the real data distribution in time, space, and activity semantics; and (ii) the category subsequence satisfies the precedence constraints encoded by $\mathbf{M}$. Unlike next-location prediction [9, 26, 41], the task outputs a *complete* time-stamped, categorized, geolocated sequence; unlike single-step conditioning, the constraint couples *multiple, non-adjacent* positions and therefore cannot be enforced by reweighting any single token independently.

Evaluation axes. We evaluate along two axes that are in tension. *Distributional fidelity* is measured by the Jensen–Shannon divergence (JSD) between generated and reference marginals of travel distance, radius of gyration, category-transition structure, daily location count, hourly category usage, and a global ranking statistic (G-RANK). *Constraint satisfaction* is measured by ordering-violation rates (OVR), pairwise and category coverage of the required relations, and an overall unsatisfied-sequence rate (Unsat); full definitions are given in Section 5. A trivial reordering can satisfy the constraint while destroying fidelity, and an unconstrained generator maximizes fidelity while ignoring the constraint; the goal is to improve constraint satisfaction with minimal fidelity cost.

4 Method

PCDG is a *training-free* control layer that is attached to a frozen, pre-trained conditional check-in generator and acts only during reverse-diffusion sampling. We first recall the backbone generator (§4.1), then formalize the partial-order constraints as differentiable energies (§4.2), describe the KL-preserving augmented-Lagrangian projection (§4.3) and the complementary distance regularizer (§4.4), and finally specify where the control is applied in the reverse process (§4.5).

4.1 Backbone: conditional joint discrete diffusion

We build on a conditional generator that models arrival times with a diffusion point process and the discrete content (categories and POIs) with a conditional discrete diffusion model, following the fine-grained trajectory generator of Deng et al. [7]. The temporal branch samples event times; the discrete branch represents a sequence as a token string $\mathbf{x} = [\text{START}, k_{1:n}, \text{SEP}, l_{1:n}, \text{END}]$, with a *category mask* and a *POI mask* identifying the two segments. Categories and POIs use separate absorbing-state transition matrices [1]; a Transformer denoiser with an $\mathbf{x}_0$ -parameterization predicts, at each reverse step, per-position logits $\mathbf{z} \in \mathbb{R}^{V \times L}$ over the joint vocabulary of size V across the L positions, from which the posterior $p(\mathbf{x}_{s-1} | \mathbf{x}_s)$ and the next discrete state are sampled. **The backbone is used as-is and is never retrained by PCDG**; our contribution modifies only the per-step sampling distribution $\mathbf{z}$.

4.2 Partial-order constraints as differentiable energies

Given the sample-level matrix $\mathbf{M}$, we parse it into a set of pairwise constraints $\{(a, b) : \mathbf{M}[a, b] > \frac{1}{2}\}$. To make a constraint differentiable in the logits, we relax the hard categorical choice at each position into a distribution. With temperature T and (optionally) Gumbel noise $\mathbf{g}_l$,

$$\pi_l = \text{softmax}((\mathbf{z}_l + \mathbf{g}_l)/T) \in \Delta^{V-1}, \quad l = 1, \dots, L. \quad (2)$$

We restrict attention to category positions $\mathcal{P}_k$ (via the category mask) and to the category slice of the vocabulary, writing $\pi_l^{\text{cat}}(a)$ for the relaxed probability that position $l \in \mathcal{P}_k$ takes category $a \in C$. For a constraint (a, b) let $P_l^A = \pi_l^{\text{cat}}(a)$ and $P_l^B = \pi_l^{\text{cat}}(b)$. We define two energies:

$$g_{(a,b)}^{\text{ord}} = \sum_{l \in \mathcal{P}_k} \left(\sum_{l' < l} P_{l'}^B \right) P_l^A, \quad (3)$$

$$g_{(a,b)}^{\text{ex}} = \text{ReLU}(1 - \sum_l P_l^A) + \text{ReLU}(1 - \sum_l P_l^B). \quad (4)$$

The *order energy* (3) is the expected number of inverted pairs, i.e. the probability mass of a b appearing *before* an a ; it is zero iff no b precedes any a . The *existence energy* (4) penalizes an expected count below one for either constrained category, which prevents the model from trivially satisfying ordering by *omitting* a or b . Both energies are computed only over category positions; POI logits do not enter Eqs. (3)–(4).

4.3 KL-preserving augmented-Lagrangian projection

At a controlled reverse step, we project the backbone logits to reduce constraint violations while staying close to the model’s own distribution. Let $\mathbf{y}^{(0)} = \mathbf{z}$ be the detached backbone logits and $\mathbf{y}$ a trainable copy. We solve, per sequence,

$$\min_{\mathbf{y}} D_{\text{KL}}(\text{softmax}(\mathbf{y}) \parallel \text{softmax}(\mathbf{y}^{(0)})) \quad \text{s.t.} \quad g_{(a,b)}^{\text{ord}} \leq \tau, \quad g_{(a,b)}^{\text{ex}} \leq \tau \quad \forall (a, b), \quad (5)$$

where the KL term anchors the projected distribution to the base model so that fidelity is preserved away from the constrained categories. We optimize the augmented-Lagrangian relaxation of (5): with $\delta^\bullet = \text{ReLU}(g^\bullet - \tau)$,

$$\begin{aligned} \mathcal{L}(\mathbf{y}) = & w_{\text{kl}} D_{\text{KL}} + \sum_{(a,b)} w_{\text{ord}} \left(\lambda^{\text{ord}} \delta^{\text{ord}} + \frac{1}{2} \mu^{\text{ord}} (\delta^{\text{ord}})^2 \right) \\ & + \sum_{(a,b)} w_{\text{ex}} \left(\lambda^{\text{ex}} \delta^{\text{ex}} + \frac{1}{2} \mu^{\text{ex}} (\delta^{\text{ex}})^2 \right). \end{aligned} \quad (6)$$

An *inner* loop takes projected-gradient steps on $\mathbf{y}$ (SGD with learning rate η , gradient-norm clipping, updates restricted to positions of constrained sequences). An *outer* loop evaluates the *hard* violations $\hat{\delta}$ (from the arg max one-hot assignment, not the relaxation) and updates the dual variables,

$$\lambda \leftarrow \lambda + \mu \hat{\delta}, \quad \mu \leftarrow \min(\alpha \mu, \mu_{\text{max}}) \text{ if } \hat{\delta} > \tau_{\text{tol}}, \quad (7)$$

stopping early once $\max \hat{\delta} < \tau_{\text{tol}}$. Evaluating the multiplier update on hard violations lets the penalty grow precisely for the sequences and constraints that remain unsatisfied, rather than applying a uniform fixed penalty. Crucially, the projection reweights the sampling distribution over category positions only; it performs *no* POI candidate filtering and imposes *no* hard category $\rightarrow$ POI consistency mask.

4.4 Distance-distribution regularization

Enforcing category order perturbs the category marginals and, indirectly, the induced spatial statistics. To preserve geographic realism we add an independent distance regularizer that matches the *population* distribution of travel length. From the training split only, we compute the cumulative great-circle (Haversine) route length d of every training sequence with at least two POIs and summarize it as a soft histogram H_{train} over $\log(1 + d)$ with fixed Gaussian bins (the reference is content-hashed and cached, never stored in the model). During projection, for each POI position we fix the top- κ candidate POIs under the backbone and draw P routes with a straight-through Gumbel-softmax (exact distance on the forward pass, a bilinear surrogate on the backward pass), forming a differentiable route-length histogram $H_{\text{proj}}(\mathbf{y})$. We add the term

$$w_{\text{dist}} D_{\text{KL}}(H_{\text{train}} \parallel H_{\text{proj}}(\mathbf{y})) \quad (8)$$

to the objective (6). Unlike the order/existence energies, this term acts on *POI-position* logits, and it is a batch-level distribution-matching regularizer rather than a per-sample hard constraint; it is applied only to sequences with at least two POIs and uses an independent random stream. Its weight is not tuned per city and is not guaranteed to improve every fidelity metric; we quantify its effect via the No Distance KL ablation (§5).

4.5 Placement in the reverse process and joint sampling

The control is applied in the *late* reverse-diffusion steps—where the predicted categories are informative enough for the energies to be meaningful—specifically during the last K steps and every f steps therein. At each such step, PCDG replaces the backbone category-position logits with their projected values (and, when enabled, the distance-adjusted POI-position logits), and the next discrete state is obtained by the backbone’s ordinary joint categorical sampling over the full token sequence. The temporal branch is unchanged. Because every operation reads and rewrites only the per-step sampling logits of a frozen model, PCDG adds no trainable parameters and requires no retraining, and it reduces to the unconstrained backbone when the projection is disabled. Algorithm pseudocode and default hyperparameters are given in Appendix A. [TODO: add Fig. 1 (framework) and Fig. 2 (one projected reverse step); export from figure_build/]

5 Experiments

We evaluate whether PCDG enforces category precedence while preserving fidelity, how it compares to retraining- and post-processing-based alternatives, and how each component contributes.

5.1 Setup

Datasets. We use check-in data from two cities, **New York** and **Istanbul**, under an out-of-distribution (OOD) split that holds out partial-order configurations unseen during training (2,108 and 4,914 held-out reference sequences, respectively). For each reference we derive a sample-level partial-order matrix $\mathbf{M}$ from the category first/last positions (§3). The base generators are trained once per city and **kept frozen** for all methods; PCDG and all sampling-time baselines reuse the identical frozen weights. [TODO: confirm final train/test counts and OOD definition from distance-v2 manifest]

Protocol. Unless noted, the constraint projection uses order weight $w_{\text{ord}}=1$, existence weight $w_{\text{ex}}=5$, token-KL weight $w_{\text{kl}}=1$, and distance-KL weight $w_{\text{dist}}=1$ (8 routes, top-32 candidates, 32 bins), with Gumbel temperature 3, applied over the last 40 reverse steps every 4 steps without early stopping. No per-city hyper-parameter search is performed. We report the mean $\pm$ sample standard deviation over **three sampling seeds** at a fixed checkpoint; the reported variability reflects sampling stochasticity, not training variability.

Baselines. We compare five methods on the same frozen backbone unless stated: **JointGen** (unconstrained backbone); **PostSwap** (generate, then stably reorder categories to satisfy $\mathbf{M}$); **EnergyGuide** (gradient guidance by the constraint-violation energy during sampling [33]); **PO-CFG** (a *retrained* classifier-free-guidance model conditioned on the partial order, trained for 1000 epochs; guidance scale 1); and **PCDG** (ours). PO-CFG is the only method that requires retraining.

Metrics. *Fidelity* uses the Jensen–Shannon divergence (JSD, lower is better) of six marginals (evaluation protocol v2): travel *Distance*, *Radius* of gyration, *CategoryTransition* (adjacent directed category pairs), *DailyLoc* (daily location count), hourly *Category* usage, and a global rank statistic *G-RANK*. *Constraint satisfaction* (percentages) uses $\text{OVR}_{\text{strict}}$ and OVR_{skip} (ordering-violation rates), *Pair coverage*

and *Category coverage* (fraction of required relations/categories realized, higher is better), and *Unsat* (fraction of sequences violating any constraint, lower is better). [TODO: add one-line formal definitions in Appendix B]

5.2 Main results

Table 2 reports the method comparison on both cities.

Findings. [TODO: fill from distance-v2 final tables] Qualitatively, we expect to report: (i) PCDG attains the strongest *constraint satisfaction* on both cities—more than halving *Unsat* relative to the unconstrained JointGen and roughly tripling *Pair coverage*—and surpasses PostSwap, EnergyGuide, and the *retrained* PO-CFG on the ordering and coverage metrics, despite being training-free; (ii) PostSwap trivially drives OVR_{skip} to 0 by construction but cannot improve coverage and distorts *Distance*; (iii) PO-CFG achieves the best raw *fidelity* on several JSD metrics, exposing a genuine fidelity–constraint trade-off that we analyze in §5.4. *All numeric claims in this subsection are to be finalized from the distance-v2 evaluation.*

5.3 Ablation study

Table 4 isolates each component by disabling it from the full model: NO PROJECTION (the backbone), NO EXISTENCE, NO ORDER, NO KL (both KL terms off), NO DISTANCE KL (only the distance term off), FIXED MULTIPLIERS (no dual update, Eq. (7)), NO GUMBEL (deterministic relaxation), and FULL (= PCDG).

Findings. [TODO: fill from distance-v2 final tables] We expect the ablation to show that the *existence* energy is responsible for coverage (removing it collapses pair/category coverage), the *order* energy is responsible for strict ordering (removing it inflates $\text{OVR}_{\text{strict}}$), and the *adaptive multipliers* materially improve constraint satisfaction over fixed penalties. The *token-KL* and *Gumbel* terms have smaller marginal effects at this projection budget, which we report honestly; the *distance-KL* term is evaluated specifically for its effect on spatial-fidelity metrics (Radius, Distance).

5.4 Fidelity–constraint trade-off

[TODO: Fig. 3: scatter of *Unsat* (x) vs. *Distance/Radius* JSD (y), one point per method per city, showing PCDG on the favorable frontier and PO-CFG trading constraint satisfaction for raw fidelity.] We position each method in the *Unsat*–fidelity plane and discuss where PCDG sits relative to the retraining-based PO-CFG and the post-hoc PostSwap.

5.5 Cost and reproducibility

PCDG adds only sampling-time projection iterations to a frozen generator and trains nothing, whereas PO-CFG requires a full retraining per city. We report wall-clock sampling overhead and the number of projection updates in Appendix C, and we release code, constraint matrices, and evaluation scripts to reproduce every table. [TODO: insert sampling-time overhead numbers from distance-v2 diagnostics]

6 Limitations

Several limitations frame our claims honestly. (i) We evaluate two cities; the Istanbul base model was trained historically with a different batch size than New York and its full data provenance is not retroactively verifiable, so we do not claim identical cross-city

Table 1: Method comparison: distributional similarity (JSD ↓). *Provisional eval-v1 anchor; final submission uses eval-v2 (CategoryTransition replaces Interval) on the distance-v2 run.*

Istanbul						
Method	Distance ↓	Radius ↓	Interval ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0049 ± 0.0005	0.0414 ± 0.0009	0.0325 ± 0.0005	0.1922 ± 0.0022
PostSwap	0.0949 ± 0.0127	0.0226 ± 0.0017	0.0049 ± 0.0005	0.0414 ± 0.0009	0.0325 ± 0.0005	0.1922 ± 0.0022
EnergyGuide	0.1156 ± 0.0052	0.0222 ± 0.0025	0.0049 ± 0.0004	0.0392 ± 0.0020	0.0342 ± 0.0004	0.1921 ± 0.0004
PO-CFG (s=1)	0.0890 ± 0.0050	0.0263 ± 0.0020	0.0049 ± 0.0005	0.0444 ± 0.0022	0.0256 ± 0.0003	0.1794 ± 0.0026
PCDG	0.1441 ± 0.0073	0.0582 ± 0.0025	0.0049 ± 0.0005	0.0366 ± 0.0008	0.0083 ± 0.0002	0.1828 ± 0.0026
New York						
Method	Distance ↓	Radius ↓	Interval ↓	DailyLoc ↓	Category ↓	G-RANK ↓
JointGen	0.2904 ± 0.0156	0.1104 ± 0.0071	0.0154 ± 0.0007	0.0409 ± 0.0035	0.0013 ± 0.0000	0.1612 ± 0.0013
PostSwap	0.2276 ± 0.0448	0.1104 ± 0.0071	0.0154 ± 0.0007	0.0409 ± 0.0035	0.0013 ± 0.0000	0.1612 ± 0.0013
EnergyGuide	0.2943 ± 0.0060	0.1112 ± 0.0061	0.0154 ± 0.0008	0.0411 ± 0.0029	0.0013 ± 0.0001	0.1622 ± 0.0009
PO-CFG (s=1)	0.1228 ± 0.0160	0.0823 ± 0.0029	0.0153 ± 0.0006	0.0499 ± 0.0008	0.0008 ± 0.0000	0.1525 ± 0.0025
PCDG	0.2484 ± 0.0267	0.1430 ± 0.0055	0.0153 ± 0.0008	0.0355 ± 0.0031	0.0047 ± 0.0008	0.1937 ± 0.0082

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city, not statistical significance.

Table 2: Method comparison: constraint satisfaction (%). PCDG attains the best ordering/coverage on both cities, surpassing the retrained PO-CFG, while being training-free. *Provisional eval-v1 anchor; to be refreshed from distance-v2.*

Istanbul					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
PostSwap	86.10 ± 0.15	0.00 ± 0.00	14.63 ± 0.19	43.19 ± 0.20	85.37 ± 0.19
EnergyGuide	90.66 ± 0.06	44.76 ± 0.58	19.17 ± 0.43	47.06 ± 0.25	92.89 ± 0.16
PO-CFG (s=1)	63.45 ± 0.23	19.09 ± 0.54	43.72 ± 0.35	66.00 ± 0.23	69.07 ± 0.45
PCDG	32.96 ± 0.46	9.10 ± 0.16	70.92 ± 0.23	85.29 ± 0.14	36.83 ± 0.24
New York					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
JointGen	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
PostSwap	78.22 ± 0.48	0.00 ± 0.00	21.13 ± 0.15	46.75 ± 0.12	78.87 ± 0.15
EnergyGuide	89.17 ± 0.14	51.59 ± 0.88	21.17 ± 0.19	47.14 ± 0.14	93.20 ± 0.39
PO-CFG (s=1)	41.91 ± 1.37	8.97 ± 0.88	59.92 ± 1.58	74.99 ± 1.08	49.48 ± 1.20
PCDG	38.48 ± 1.04	13.76 ± 0.72	67.33 ± 1.86	81.38 ± 0.81	42.77 ± 1.79

Mean ± sample standard deviation over three sampling seeds. Bold denotes the best unrounded mean within each city. PostSwap drives OVR_skip to 0 by construction but cannot improve coverage.

training conditions. (ii) The reported error bars are sample standard deviations over three sampling seeds at a *fixed* checkpoint; they reflect sampling stochasticity, not training variability. (iii) PO-CFG is evaluated at guidance scale 1; we do not claim an extrapolation-tuned optimum. (iv) The distance-KL weight is fixed, not tuned per city, and is a batch-level distribution-matching regularizer rather than a per-sample spatial guarantee. (v) PCDG reduces—but does not certify zero—constraint violations; it reweights the sampling distribution and performs no hard candidate filtering. (vi) We do not yet report a downstream-utility study (e.g., training a next-POI model on the synthetic data). Addressing complex partial-order DAGs (transitivity, conflict detection), adaptive constraint scheduling, a training-time partial-order objective, and downstream utility are natural directions for future work.

7 Conclusion

We studied controllable synthetic generation of POI check-in sequences under sample-level activity-category precedence constraints—a web-native, privacy-relevant problem that single-step conditioning cannot address. We introduced PCDG, a training-free, sampling-time projection that compiles a partial-order matrix into differentiable order and existence energies, projects the category-position logits of a frozen joint temporal + category + POI discrete-diffusion generator via a KL-preserving augmented Lagrangian, and preserves spatial fidelity through a distance-distribution regularizer on POI positions. Across two cities, PCDG delivers the strongest constraint satisfaction among post-hoc, guidance, and even fully retrained conditional baselines, at a characterized and modest fidelity cost, without retraining the generator. We hope the training-free, sampling-time projection recipe proves useful for controllable generation of other structured web content.

A Algorithm and Default Hyper-parameters

Algorithm 1 summarizes one projected reverse step. Defaults: $w_{\text{ord}}=1$, $w_{\text{ex}}=5$, $w_{\text{kl}}=1$, $w_{\text{dist}}=1$; Gumbel temperature $T=3$; outer iterations

Table 3: Ablation: distributional similarity (JSD ↓). *Provisional eval-v1 anchor; distance-v2 adds a No DISTANCE KL row and uses eval-v2.*

Istanbul						
Method	Distance ↓	Radius ↓	Interval ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.1128 ± 0.0087	0.0226 ± 0.0017	0.0049 ± 0.0005	0.0414 ± 0.0009	0.0325 ± 0.0005	0.1922 ± 0.0022
No Existence	0.1134 ± 0.0105	0.0220 ± 0.0030	0.0049 ± 0.0005	0.0418 ± 0.0021	0.0336 ± 0.0006	0.1923 ± 0.0021
No Order	0.1232 ± 0.0106	0.0538 ± 0.0032	0.0049 ± 0.0005	0.0357 ± 0.0008	0.0066 ± 0.0004	0.1756 ± 0.0037
No KL	0.1406 ± 0.0117	0.0574 ± 0.0028	0.0049 ± 0.0005	0.0361 ± 0.0012	0.0083 ± 0.0003	0.1817 ± 0.0024
Fixed Multipliers	0.1425 ± 0.0067	0.0579 ± 0.0032	0.0049 ± 0.0004	0.0366 ± 0.0008	0.0082 ± 0.0003	0.1837 ± 0.0017
No Gumbel Noise	0.1396 ± 0.0088	0.0586 ± 0.0023	0.0049 ± 0.0005	0.0363 ± 0.0005	0.0082 ± 0.0003	0.1822 ± 0.0035
Full	0.1441 ± 0.0073	0.0582 ± 0.0025	0.0049 ± 0.0005	0.0366 ± 0.0008	0.0083 ± 0.0002	0.1828 ± 0.0026
New York						
Method	Distance ↓	Radius ↓	Interval ↓	DailyLoc ↓	Category ↓	G-RANK ↓
No Projection	0.2904 ± 0.0156	0.1104 ± 0.0071	0.0154 ± 0.0007	0.0409 ± 0.0035	0.0013 ± 0.0000	0.1612 ± 0.0013
No Existence	0.2671 ± 0.0392	0.1085 ± 0.0067	0.0154 ± 0.0007	0.0395 ± 0.0042	0.0016 ± 0.0000	0.1585 ± 0.0008
No Order	0.2332 ± 0.0532	0.1343 ± 0.0108	0.0154 ± 0.0008	0.0346 ± 0.0026	0.0026 ± 0.0004	0.1867 ± 0.0047
No KL	0.2415 ± 0.0239	0.1421 ± 0.0046	0.0153 ± 0.0008	0.0352 ± 0.0041	0.0047 ± 0.0008	0.1939 ± 0.0064
Fixed Multipliers	0.2562 ± 0.0555	0.1388 ± 0.0064	0.0153 ± 0.0008	0.0351 ± 0.0032	0.0036 ± 0.0005	0.1895 ± 0.0013
No Gumbel Noise	0.2302 ± 0.0068	0.1431 ± 0.0086	0.0154 ± 0.0008	0.0346 ± 0.0052	0.0046 ± 0.0007	0.1917 ± 0.0028
Full	0.2484 ± 0.0267	0.1430 ± 0.0055	0.0153 ± 0.0008	0.0355 ± 0.0031	0.0047 ± 0.0008	0.1937 ± 0.0082

Table 4: Ablation: constraint satisfaction (%). Existence energy drives coverage; order energy drives strict ordering; adaptive multipliers beat fixed penalties. *Provisional eval-v1 anchor; distance-v2 adds No DISTANCE KL.*

Istanbul					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	92.89 ± 0.08	48.05 ± 0.74	14.63 ± 0.19	43.19 ± 0.20	95.57 ± 0.15
No Existence	93.68 ± 0.18	7.49 ± 0.61	7.32 ± 0.16	36.90 ± 0.22	93.58 ± 0.20
No Order	63.52 ± 0.69	50.41 ± 0.68	72.64 ± 0.71	86.12 ± 0.19	70.64 ± 0.52
No KL	32.85 ± 0.52	9.20 ± 0.29	71.11 ± 0.08	85.40 ± 0.16	36.75 ± 0.57
Fixed Multipliers	42.73 ± 0.52	20.96 ± 0.77	71.38 ± 0.82	85.05 ± 0.41	45.58 ± 0.58
No Gumbel Noise	32.66 ± 0.46	9.01 ± 0.31	71.14 ± 0.51	85.42 ± 0.28	36.41 ± 0.17
Full	32.96 ± 0.46	9.10 ± 0.16	70.92 ± 0.23	85.29 ± 0.14	36.83 ± 0.24
New York					
Method	OVR_strict ↓	OVR_skip ↓	Pair cov. ↑	Category cov. ↑	Unsat ↓
No Projection	89.45 ± 0.20	52.92 ± 0.86	21.13 ± 0.15	46.75 ± 0.12	93.64 ± 0.26
No Existence	90.28 ± 0.50	17.92 ± 1.75	11.01 ± 0.51	39.48 ± 0.12	91.72 ± 0.41
No Order	65.26 ± 0.28	51.26 ± 0.15	69.80 ± 0.91	82.83 ± 0.32	72.92 ± 0.85
No KL	38.24 ± 0.96	13.98 ± 0.82	67.67 ± 1.56	81.57 ± 0.72	42.40 ± 1.48
Fixed Multipliers	58.22 ± 0.56	35.19 ± 0.36	64.08 ± 0.61	78.53 ± 0.25	60.46 ± 0.41
No Gumbel Noise	38.33 ± 1.43	13.95 ± 1.21	67.36 ± 0.96	81.47 ± 0.71	42.86 ± 1.38
Full	38.48 ± 1.04	13.76 ± 0.72	67.33 ± 1.86	81.38 ± 0.81	42.77 ± 1.79

10, inner iterations 50; SGD learning rate η , gradient clip 10; multiplier growth $\alpha=2$, $\mu_{\max}=10^3$, tolerance $\tau_{\text{tol}}=0.25$; distance routes $P=8$, candidates $\kappa=32$, bins 32; applied over the last 40 reverse steps every 4 steps. [TODO: confirm η and step schedule against the distance-v2 manifest before camera-ready.]

B Metric Definitions

[TODO: one-line formal definitions of OVR_strict, OVR_skip, Pair/Category coverage, Unsat, and the six eval-v2 JSD marginals; copy from the unified evaluation module.]

C Cost

[TODO: sampling wall-clock overhead vs. JointGen, and PO-CFG retraining cost (per-city GPU-hours), from distance-v2 diagnostics.]

References

- [1] Jacob Austin, Daniel Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. 2021. Structured Denoising Diffusion Models in Discrete State-Spaces. In *Advances in Neural Information Processing Systems*. arXiv preprint arXiv:2107.03006; NeurIPS 2021.
- [2] Chu Cao and Mo Li. 2021. Generating Mobility Trajectories with Retained Data Utility. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3447548.3467158
- [3] Michael Cardei, Jacob K. Christopher, Thomas Hartvigsen, Bhavya Kailkhura, and Ferdinando Fioretto. 2025. Constrained Discrete Diffusion. In *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*. doi:10.52202/085713-0415 39th Conference on Neural Information Processing Systems (NeurIPS 2025).
- [4] C. Chen, H. Zhang, P. Wang, and F. Lu. 2024. Simulating Human Mobility with a Trajectory Generation Framework Based on Diffusion Model. *International Journal of Geographical Information Science* (2024). doi:10.1080/13658816.2024.2312199
- [5] Jinghong Chen, Weizhe Lin, Jingbiao Mei, and Bill Byrne. 2024. Control-DAG: Constrained Decoding for Non-Autoregressive Directed Acyclic T5 using Weighted Finite State Automata. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. doi:10.18653/v1/2024.naacl-short.42

Algorithm 1 PCDG projected reverse step (one controlled step s)

Require: backbone logits $\mathbf{z}$; constraints $\{(a, b)\}$ with $\mathbf{W}_A, \mathbf{W}_B$; masks; distance reference H_{train}

- 1: $\mathbf{y} \leftarrow \mathbf{z}$ (detached); initialize λ, μ for active rows
- 2: **for** outer = 1 to T_{out} **do**
- 3: **for** inner = 1 to T_{in} **do**
- 4: relax $\mathbf{y}$ to π ; compute $g^{\text{ord}}, g^{\text{ex}}$ on category positions $\triangleright$ Eq. (3)–(4)
- 5: $\mathcal{L} \leftarrow w_{\text{kl}} D_{\text{KL}} + \text{AL}(g^{\text{ord}}, g^{\text{ex}}; \lambda, \mu) + w_{\text{dist}} D_{\text{KL}}(H_{\text{train}} \| H_{\text{proj}})$
- 6: SGD step on $\mathbf{y}$ (clip; restrict to active rows)
- 7: **end for**
- 8: evaluate hard violations $\hat{\delta}$; update $\lambda, \mu \triangleright$ Eq. (7); early stop if satisfied
- 9: **end for**
- 10: **return** joint categorical sample from projected $\mathbf{y}$

- [6] Jialei Chen, Yuanbo Xu, and Yiheng Jiang. 2025. Unlocking the Power of Diffusion Models in Sequential Recommendation: A Simple and Effective Approach. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3737172
- [7] Bangchao Deng, Ling Ding, Lianhua Ji, Chunhua Chen, Xin Jing, Bingqing Qu, and Dingqi Yang. 2025. Marionette: Fine-Grained Conditional Generative Modeling of Spatiotemporal Human Trajectory Data Beyond Imitation. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3737039
- [8] Bangchao Deng, Xin Jing, Tianyue Yang, Bingqing Qu, Dingqi Yang, and Philippe Cudre-Mauroux. 2025. Revisiting Synthetic Human Trajectories: Imitative Generation and Benchmarks Beyond Datasaurus. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3690624.3709180
- [9] Jie Feng, Yong Li, Chao Zhang, Funing Sun, Fanchao Meng, Ang Guo, and Depeng Jin. 2018. DeepMove: Predicting Human Mobility with Attentional Recurrent Networks. In *Proceedings of the 2018 World Wide Web Conference*. doi:10.1145/3178876.3186058
- [10] Jie Feng, Zeyu Yang, Fengli Xu, Haisu Yu, Mudan Wang, and Yong Li. 2020. Learning to Simulate Human Mobility. In *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3394486.3412862
- [11] Saibo Geng, Martin Josifoski, Maxime Peyrard, and Robert West. 2023. Grammar-Constrained Decoding for Structured NLP Tasks without Finetuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. doi:10.18653/v1/2023.emnlp-main.674
- [12] Letian Gong, Shengnan Guo, Yan Lin, Yichen Liu, Erwen Zheng, Yiwei Shuang, Youfang Lin, Jilin Hu, and Huaiyu Wan. 2025. STCDM: Spatio-Temporal Contrastive Diffusion Model for Check-In Sequence Generation. *IEEE Transactions on Knowledge and Data Engineering* 37, 4 (2025). doi:10.1109/TKDE.2025.3525718
- [13] Marta C. González, César A. Hidalgo, and Albert-László Barabási. 2008. Understanding Individual Human Mobility Patterns. *Nature* (2008). doi:10.1038/nature07850
- [14] A. Haydari, D. Chen, Z. Lai, et al. 2026. MobilityGPT: Enhanced Human Mobility Modeling With a GPT Model. *IEEE Transactions on Intelligent Transportation Systems* (2026). doi:10.1109/TITS.2025.3626357
- [15] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems*. arXiv preprint arXiv:2006.11239; NeurIPS 2020.
- [16] S. L. Hsu, E. Tung, J. Krumm, et al. 2024. TrajGPT: Controlled Synthetic Trajectory Generation Using a Multitask Transformer-Based Spatiotemporal Model. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. doi:10.1145/3678717.3691303
- [17] Y. Hu, Y. Du, Z. Zhang, et al. 2024. Real-Time Trajectory Synthesis with Local Differential Privacy. In *Proceedings of the IEEE International Conference on Data Engineering*. doi:10.1109/ICDE60146.2024.00137
- [18] Yanmei Hu, Minghao Yin, and Dunbo Cai. 2011. On the Discovery and Utility of Precedence Constraints in Temporal Planning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 25. doi:10.1609/aaai.v25i1.8052
- [19] W. Jiang, W. X. Zhao, J. Wang, and J. Jiang. 2023. Continuous Trajectory Generation Based on Two-Stage GAN. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. doi:10.1609/aaai.v37i4.25557
- [20] S. Li, T. Tran, H. Lin, et al. 2025. Geo-Llama: Leveraging LLMs for Human Mobility Trajectory Generation with Constraints. In *Proceedings of the 2025 IEEE International Conference on Mobile Data Management*. doi:10.1109/MDM65600.2025.00023
- [21] H. Lin, S. Shaham, Y. Y. Chiang, and C. Shahabi. 2023. Generating Realistic and Representative Trajectories with Mobility Behavior Clustering. In *Proceedings of the 31st ACM International Conference on Advances in Geographic Information Systems*. doi:10.1145/3589132.3625657
- [22] Jing Long, Guanhua Ye, Tong Chen, Yang Wang, Meng Wang, and Hongzhi Yin. 2024. Diffusion-Based Cloud-Edge-Device Collaborative Learning for Next POI Recommendations. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3637528.3671743
- [23] Q. Long, C. Rong, T. Li, and Y. Li. 2026. Dynamic Population Distribution Aware Human Trajectory Generation with Diffusion Model. *ACM Transactions on Intelligent Systems and Technology* (2026). doi:10.1145/3787459
- [24] Qingyue Long, Huandong Wang, Tong Li, Lisi Huang, Kun Wang, Qiong Wu, Guangyu Li, Yanping Liang, Li Yu, and Yong Li. 2023. Practical Synthetic Human Trajectories Generation Based on Variational Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3580305.3599888
- [25] Aaron Lou, Chenlin Meng, and Stefano Ermon. 2024. Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution. In *Proceedings of the 41st International Conference on Machine Learning*. arXiv preprint arXiv:2310.16834; ICML 2024 Oral.
- [26] Yingtao Luo, Qiang Liu, and Zhaocheng Liu. 2021. STAN: Spatio-Temporal Attention Network for Next Location Recommendation. In *Proceedings of the Web Conference 2021*. doi:10.1145/3442381.3449998
- [27] Chenlin Meng, Kristy Choi, Jiaming Song, and Stefano Ermon. 2022. Concrete Score Matching: Generalized Score Matching for Discrete Data. In *Advances in Neural Information Processing Systems*, Vol. 35. doi:10.52202/068431-2502
- [28] F. Netzer and M. Lienkamp. 2024. Privacy Preserving Human Mobility Generation Using Grid-Based Data and Graph Autoencoders. *ISPRS International Journal of Geo-Information* 13, 7 (2024). doi:10.3390/ijgi13070245
- [29] Y. Qin, H. Wu, W. Ju, et al. 2023. A Diffusion Model for POI Recommendation. *ACM Transactions on Information Systems* (2023). doi:10.1145/3624475
- [30] J. Rao, S. Gao, and S. Zhu. 2023. CATS: Conditional Adversarial Trajectory Synthesis for Privacy-Preserving Trajectory Data Publication Using Deep Learning Approaches. *International Journal of Geographical Information Science* (2023). doi:10.1080/13658816.2023.2262550
- [31] X. Rao, S. Shang, R. Jiang, et al. 2025. Seed: Bridging Sequence and Diffusion Models for Road Trajectory Generation. In *Proceedings of the ACM Web Conference 2025*. doi:10.1145/3696410.3714951
- [32] Subham Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin Chiu, Alexander Rush, and Volodymyr Kuleshov. 2024. Simple and Effective Masked Diffusion Language Models. In *Advances in Neural Information Processing Systems*, Vol. 37. doi:10.52202/079017-4135
- [33] Yair Schiff, Subham Sekhar Sahoo, Hao Phung, Guanghan Wang, Sam Boshar, Hugo Dalla-Torre, Bernardo P. de Almeida, Alexander Rush, Thomas Pierrot, and Volodymyr Kuleshov. 2025. Simple Guidance Mechanisms for Discrete Diffusion Models. In *International Conference on Learning Representations*. arXiv preprint arXiv:2412.10193; ICLR 2025.
- [34] Chaoming Song, Zehui Qu, Nicholas Blumm, and Albert-László Barabási. 2010. Limits of Predictability in Human Mobility. *Science* (2010). doi:10.1126/science.1177170
- [35] X. Sun, Q. Ye, H. Hu, et al. 2023. Synthesizing Realistic Trajectory Data With Differential Privacy. *IEEE Transactions on Intelligent Transportation Systems* (2023). doi:10.1109/TITS.2023.3241290
- [36] D. Wang, Y. Huang, S. Gao, et al. 2025. Generative Next POI Recommendation with Semantic ID. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3711896.3736981
- [37] H. Wang, C. Gao, Y. Wu, et al. 2023. PateGail: A Privacy-Preserving Mobility Trajectory Generator with Imitation Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. doi:10.1609/aaai.v37i12.26700
- [38] Ziwei Wang, Jun Zeng, Lin Zhong, Ling Liu, Min Gao, and Junhao Wen. 2024. DS-DRec: Next POI Recommendation Using Deep Semantic Extraction and Diffusion Model. *Information Sciences* (2024). doi:10.1016/j.ins.2024.121004
- [39] T. Wei, Y. Lin, S. Guo, et al. 2024. Diff-RNTraj: A Structure-Aware Diffusion Model for Road Network-Constrained Trajectory Generation. *IEEE Transactions on Knowledge and Data Engineering* (2024). doi:10.1109/TKDE.2024.3460051
- [40] G. Xiong, Z. Li, M. Zhao, et al. 2024. TrajSGAN: A Semantic-Guiding Adversarial Network for Urban Trajectory Generation. *IEEE Transactions on Computational Social Systems* (2024). doi:10.1109/TCSS.2023.3235923
- [41] Song Yang, Jiamou Liu, and Kaiqi Zhao. 2022. GETNext: Trajectory Flow Map Enhanced Transformer for Next POI Recommendation. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. doi:10.1145/3477495.3531983

- [42] Yuan Yuan, Jingtao Ding, Chenyang Shao, Depeng Jin, and Yong Li. 2023. Spatio-temporal Diffusion Point Processes. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3580305.3599511
- [43] Yuan Yuan, Jingtao Ding, Huandong Wang, Depeng Jin, and Yong Li. 2022. Activity Trajectory Generation via Modeling Spatiotemporal Dynamics. In *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3534678.3542671
- [44] K. Zhang, Y. Pang, Y. Zhang, and Y. Sekimoto. 2024. MobGLM: A Large Language Model for Synthetic Human Mobility Generation. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. doi:10.1145/3678717.3691311
- [45] Yuanshao Zhu, Yongchao Ye, Shiyao Zhang, Xiangyu Zhao, and James J. Q. Yu. 2023. DiffTraj: Generating GPS Trajectory with Diffusion Probabilistic Model. In *Advances in Neural Information Processing Systems*, Vol. 36. doi:10.52202/075280-2843
- [46] Yuanshao Zhu, James Jianqiao Yu, Xiangyu Zhao, Qidong Liu, Yongchao Ye, Wei Chen, Zijian Zhang, Xuetao Wei, and Yuxuan Liang. 2024. ControlTraj: Controllable Trajectory Generation with Topology-Constrained Diffusion Model. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. doi:10.1145/3637528.3671866